

Introduction to logistic regression

Warmup

Work on the warmup activity (handout).

Warmup

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

What is the estimated odds of acceptance for a student with a GPA of 4.0?

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45(4) \\ = 2.59$$

$$\text{odds} = \frac{\hat{\pi}}{1 - \hat{\pi}} = e^{\{2.59\}} = 13.33$$

Warmup

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

What is the estimated probability of acceptance for a student with a GPA of 4.0?

$$\begin{aligned} \underbrace{\frac{\hat{\pi}}{1 - \hat{\pi}}}_{\text{odds}} &= 13.33 \Rightarrow \hat{\pi} = 13.33(1 - \hat{\pi}) \\ \Rightarrow \hat{\pi} &= \frac{13.33}{1 + 13.33} = 0.43 \\ \hat{\pi} &= \frac{\text{odds}}{1 + \text{odds}} \\ \hat{\pi} &= 13.33 - 13.33\hat{\pi} \\ \hat{\pi} + 13.33\hat{\pi} &= 13.33 \\ \hat{\pi}(1 + 13.33) &= 13.33 \end{aligned}$$

Warmup

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 4.0 to a student with a GPA of 3.0.

$$\frac{\exp\{-19.21 + 5.45(4)\}}{\exp\{-19.21 + 5.45(3)\}} = 232.76$$

Warmup

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 4.0 to a student with a GPA of 3.0.

$$\frac{\exp\{-19.21 + 5.45 \cdot 4\}}{\exp\{-19.21 + 5.45 \cdot 3\}} = 232.76$$

So the odds of acceptance are 232.76 times higher for a student with a GPA of 4.0, compared to a student with a GPA of 3.0

Warmup

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 3.8 to a student with a GPA of 2.8.

$$\frac{\exp\{-19.21 + 5.45(3.8)\}}{\exp\{-19.21 + 5.45(2.8)\}} = 232.76$$

Warmup

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

Calculate the odds ratio comparing odds of acceptance for a student with a GPA of 3.8 to a student with a GPA of 2.8.

$$\frac{\exp\{-19.21 + 5.45 \cdot 3.8\}}{\exp\{-19.21 + 5.45 \cdot 2.8\}} = 232.76$$

So the odds of acceptance are 232.76 times higher for a student with a GPA of 3.8, compared to a student with a GPA of 2.8

This is the same odds ratio as comparing a 4.0 GPA to a 3.0 GPA!

Warmup

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

Odds ratio for increasing GPA by one point:

$$\frac{\text{odds when GPA} = x+1}{\text{odds when GPA} = x} = \frac{\exp\{-19.21 + 5.45(x+1)\}}{\exp\{-19.21 + 5.45x\}}$$

$$= \frac{e^{-19.21} e^{5.45x} e^{5.45}}{e^{-19.21} e^{5.45x}} = e^{5.45} = 232.76$$

A one-unit increase in GPA is associated with a change in odds of acceptance by a factor of $e^{5.45} = 232.76$

Interpreting coefficients

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = \hat{\beta}_0 + \hat{\beta}_1 x$$

Interpretation of $\hat{\beta}_1$:

- + A one unit increase in x is associated with an increase of $\hat{\beta}_1$ in the log odds
- + A one unit increase in x is associated with an increase in the odds by a factor of $\exp\{\hat{\beta}_1\}$

How do you think we interpret $\hat{\beta}_0$?

Interpreting coefficients

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = \hat{\beta}_0 + \hat{\beta}_1 x$$

Interpretation of $\hat{\beta}_1$:

- + A one unit increase in x is associated with an increase of $\hat{\beta}_1$ in the log odds
- + A one unit increase in x is associated with an increase in the odds by a factor of $\exp\{\hat{\beta}_1\}$

Interpretation of $\hat{\beta}_0$:

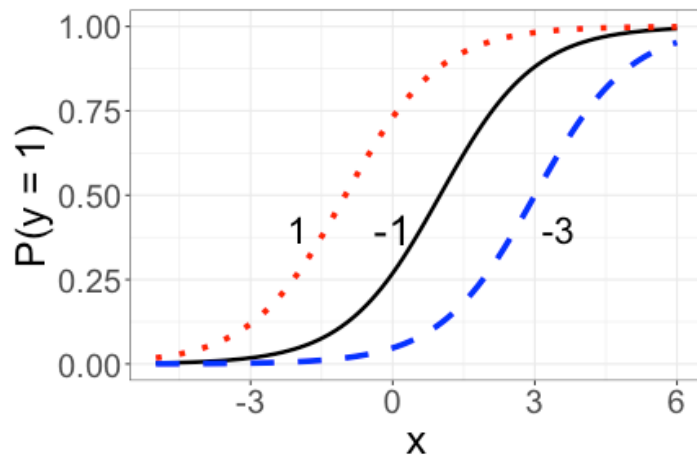
- + The estimated log odds when $x = 0$ are $\hat{\beta}_0$
- + The estimated odds when $x = 0$ are $\exp\{\hat{\beta}_0\}$

Shape of the fitted curve

$$\hat{\pi} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x}}$$

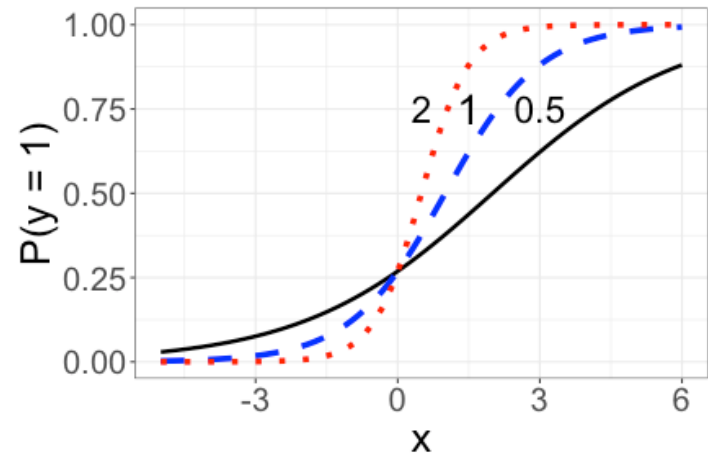
$$\hat{\pi} = \frac{\exp\{\hat{\beta}_0 + x\}}{1 + \exp\{\hat{\beta}_0 + x\}}$$

$$\hat{\beta}_0 = -3, -1, 1$$



$$\hat{\pi} = \frac{\exp\{-1 + \hat{\beta}_1 x\}}{1 + \exp\{-1 + \hat{\beta}_1 x\}}$$

$$\hat{\beta}_1 = 0.5, 1, 2$$



Fitting logistic regression in R

```
library(Stat2Data)
data("MedGPA")
med_glm <- glm(Acceptance ~ GPA, data = MedGPA,
               family = binomial)
```

response *explanatory*

↙ ↘

- + glm stands for "Generalized Linear Model"
- + Use `family = binomial` for binary responses

Fitting logistic regression in R

```
library(Stat2Data)
```

```
data("MedGPA")
```

```
med_glm <- glm(Acceptance ~ GPA, data = MedGPA,  
               family = binomial)
```

```
summary(med_glm)
```

z, not t, for logistic regression

	Estimate	Std. Error	<i>z</i> value	Pr(> <i>z</i>)	
(Intercept)	-19.207	5.629	-3.412	0.000644	***
GPA	5.454	1.579	3.454	0.000553	***

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -19.21 + 5.45 \text{ GPA}$$

Class activity

Work on the class activity on the course website. Submit your HTML on Canvas at the end of class.

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

What is the change in the odds of acceptance associated with an increase of 1 point on the MCAT?

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

What is the change in the odds of acceptance associated with an increase of 1 point on the MCAT?

Answer: An increase of 1 point on the MCAT is associated with an increase in the odds of acceptance by a factor of $\exp\{0.246\} = 1.28$.

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

What is the estimated probability that a student with an MCAT score of 40 is accepted?

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

What is the estimated probability that a student with an MCAT score of 40 is accepted?

$$\text{Answer: } \hat{\pi} = \frac{\exp\{-8.712 + 0.246 \cdot 40\}}{1 + \exp\{-8.712 + 0.246 \cdot 40\}} = 0.76$$

So we estimate that a student with an MCAT score of 40 has a 76% chance of being accepted to medical school.

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

For approximately what MCAT score would a student have a roughly 50-50 chance of being accepted to medical school?

Class activity

$$\pi = P(\textit{Accepted} = 1)$$

$$\log\left(\frac{\hat{\pi}}{1 - \hat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

For approximately what MCAT score would a student have a roughly 50-50 chance of being accepted to medical school?

Answer: We need odds = 1, so

$$\frac{\hat{\pi}}{1 - \hat{\pi}} = \exp\{-8.712 + 0.246 \text{ MCAT}\} = 1$$

So $-8.712 + 0.246 \text{ MCAT} = 0$, which means
 $\text{MCAT} = 8.712/0.246 = 35.4$